Abstract Algebra III: Assignment - Week 13

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May 23, 2026

Problem 1

Write out a proof of Maschke's Theorem in the case of representations over $\mathbb{C}$ along the following lines.

1. Given a representation
$$\rho : G \to GL(V),$$
where V is a vector space over $\mathbb{C}$, let $(\cdot,\cdot)$ be any positive definite Hermitian form on V. Define a new form $(\cdot,\cdot)_1$ on V by
$$(v,w)_1 = \frac{1}{|G|}\sum_{g\in G}(gv,gw).$$
Show that $(\cdot,\cdot)_1$ is a positive definite Hermitian form, preserved under the action of G; that is,
$$(v,w)_1 = (gv,gw)_1$$
always.

2. If W is a subrepresentation of V, show that
$$V = W \oplus W^{\perp}$$
as representations.

Solution: (1) For $a,b\in\mathbb{C}$,
$$(av_1+bv_2,w)_1 = \frac{1}{|G|}\sum_{g\in G}(a\,gv_1+b\,gv_2,gw) = a(v_1,w)_1 + b(v_2,w)_1,$$
and similarly the second variable is conjugate-linear. Moreover,
$$(w,v)_1 = \frac{1}{|G|}\sum_{g\in G}(gw,gv) = \overline{\frac{1}{|G|}\sum_{g\in G}(gv,gw)} = \overline{(v,w)_1}.$$
Thus $(\cdot,\cdot)_1$ is Hermitian. Next, for $v\neq 0$, since each $g\in G$ acts invertibly, we have $gv\neq 0$. Hence
$$(v,v)_1 = \frac{1}{|G|}\sum_{g\in G}(gv,gv) > 0.$$

Therefore $(\cdot,\cdot)_1$ is positive definite. Finally, we show that this form is preserved by the action of G. Let $h \in G$. Then

$$(hv, hw)_1 = \frac{1}{|G|}\sum_{g\in G}(ghv, ghw).$$

As g runs over G, so does gh. Therefore

$$(hv, hw)_1 = \frac{1}{|G|}\sum_{u\in G}(uv, uw) = (v, w)_1.$$

Thus

$$(v, w)_1 = (hv, hw)_1$$

for all $h \in G$, so $(\cdot,\cdot)_1$ is G-invariant.

(2) Choose a basis of V, so that $V \simeq \mathbb{C}^n$, and take the standard positive definite Hermitian form

$$(v, w) = \sum_i v_i \overline{w_i}.$$

Averaging this form over the G-action gives a new Hermitian form

$$(v, w)_1 = \frac{1}{|G|}\sum_{g\in G}(gv, gw).$$

By the previous part, $(\cdot,\cdot)_1$ is positive definite and G-invariant.

Now let $W \subset V$ be a subrepresentation, and define

$$W^\perp = \{v \in V \mid (v, w)_1 = 0 \text{ for all } w \in W\}.$$

Since $(\cdot,\cdot)_1$ is positive definite, we have

$$V = W \oplus W^\perp$$

as vector spaces.

It remains to show that $W^\perp$ is G-stable. Let $u \in W^\perp$, $h \in G$, and $w \in W$. Since W is a subrepresentation, $h^{-1}w \in W$. Using G-invariance of $(\cdot,\cdot)_1$, we get

$$(hu, w)_1 = (u, h^{-1}w)_1 = 0.$$

Hence $hu \in W^\perp$, so $W^\perp$ is a subrepresentation. Therefore

$$V = W \oplus W^\perp$$

as representations. ∎

Problem 2

Prove the following theorem of Burnside: let G be a finite group, k an algebraically closed field in which $|G|$ is invertible, and

$$\rho : G \to GL(V)$$

be a representation over k. By taking a basis of V, write each endomorphism $\rho(g)$ as a matrix. Let $\dim V = n$. Show that the representation is simple if and only if there exist n^2 elements $g_1, \ldots, g_{n^2}$ of G so that the matrices

$$\rho(g_1), \ldots, \rho(g_{n^2})$$

are linearly independent, and that this happens if and only if the algebra homomorphism

$$kG \to \mathrm{End}_k(V)$$

is surjective. Note that ρ itself is generally not surjective.

Solution: Let

$$A = \mathrm{span}_k\{\rho(g) : g \in G\} \subseteq \mathrm{End}_k(V).$$

Then A is the image of the algebra homomorphism $kG \to \mathrm{End}_k(V)$.

Suppose that there exist $g_1, \ldots, g_{n^2} \in G$ such that $\rho(g_1), \ldots, \rho(g_{n^2})$ are linearly independent. Since $\dim_k \mathrm{End}_k(V) = n^2$, they form a basis of $\mathrm{End}_k(V)$. Hence every elementary matrix E_{ij}, sending e_j to e_i and all other basis vectors to 0, can be written as a k-linear combination of the matrices $\rho(g)$. Let $0 \neq W \subseteq V$ be a subrepresentation and choose $0 \neq v = \sum_{\ell=1}^{n} c_\ell e_\ell \in W$. Pick j with $c_j \neq 0$. Then for every i,

$$\frac{1}{c_j} E_{ij} v = e_i.$$

Since W is stable under every $\rho(g)$, it is stable under their linear combinations. Thus $e_i \in W$ for all i, so $W = V$. Therefore V is simple.

Conversely, suppose that V is simple. Let

$$A = \mathrm{im}(kG \to \mathrm{End}_k(V)).$$

Since $|G|$ is invertible in k, Maschke's theorem implies that kG is semisimple. Hence its quotient A is also semisimple.

Since V is simple as a G-representation, it is simple as an A-module. By Schur's lemma, because k is algebraically closed,

$$\mathrm{End}_A(V) = k.$$

By the Artin–Wedderburn theorem, A is a product of matrix algebras over k. Since V is simple, only one simple factor acts nontrivially on V. But A is defined as the image in $\mathrm{End}_k(V)$, so its action on V is faithful. Therefore A has only this one factor, and its action on V identifies it with

$$\mathrm{End}_k(V).$$

Thus

$$A = \mathrm{End}_k(V).$$

Hence $\dim_k A = n^2$. Since A is spanned by the matrices $\rho(g)$, one can choose n^2 linearly independent matrices among them. ■

Problem 3

Let

$$G = \langle x \mid x^n = 1 \rangle$$

be a cyclic group of order n, and let $\zeta_n \in \mathbb{C}$ be a primitive n-th root of unity. Then the simple complex characters of G are the n functions

$$\chi_r(x^s) = \zeta_n^{rs},$$

where $0 \leq r \leq n-1$.

Solution: Let V be a simple complex representation of G, and set $T = \rho(x)$. Since $x^n = 1$, we have $T^n = I$. Hence the minimal polynomial of T divides $t^n - 1$. Over $\mathbb{C}$, the polynomial $t^n - 1 = \prod_{r=0}^{n-1}(t - \zeta_n^r)$ has distinct roots, so T is diagonalizable and its eigenvalues are among $\zeta_n^0, \dots, \zeta_n^{n-1}$.

For an eigenvalue λ, the eigenspace $V_\lambda = \{v \in V \mid Tv = \lambda v\}$ is G-stable, because $G = \langle x \rangle$. Since V is simple, only one eigenspace can be nonzero. Thus $V = V_\lambda$ for some λ, so $\dim V = 1$.

Therefore every simple complex representation of G is one-dimensional. Since $\rho(x)^n = 1$, we have $\rho(x) = \zeta_n^r$ for some $0 \leq r \leq n-1$. Hence, for every s,

$$\chi_r(x^s) = \rho(x^s) = \rho(x)^s = (\zeta_n^r)^s = \zeta_n^{rs}.$$

These n characters are distinct since $\chi_r(x) = \zeta_n^r$. Hence they are exactly all the simple complex characters of G. ■